

INDIAN STATISTICAL INSTITUTE, KOLKATA

Midterm , Second semester 2024-25

Algebra 2 , M.Math 1

Full Marks : 30

Time : 2 hours

Date : 26/02/2025

1. Consider the action of D_n on the set of vertices of a regular n -gon. A subgroup $G \subset D_n$ is called transitive if the induced action of G on the set of vertices of the regular n -gon is transitive.

(a) Find a transitive subgroup $G \subset D_n$ such that $G \neq D_n$. [2]

(b) Show that for n odd a transitive subgroup $G \subset D_n$ is equal to D_n if and only if G contains a reflection. [4]

(c) Find a transitive subgroup $G \subset D_4$ such that G contains a reflection but $G \neq D_4$. [3]

(d) Show that for n -even a transitive subgroup $G \subset D_n$ is equal to D_n if and only if G contains a reflection which fixes a vertex. [3]

2. Let G be a group of order 351. then G has normal Sylow p -subgroup for some prime p dividing 351. [4]

3. Find the degree and a basis of the following extension L/K (justify your answer):

(a) $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{2}, i)$. [4]

(b) $K = \mathbb{R}(x + \frac{1}{x})$, $L = \mathbb{R}(x)$. [3]

4. Let k be a field, $f(x) \in k[x]$ be an irreducible polynomial, and let L/k be a splitting field of $f(x)$ over k . Let

$$\text{Aut}_k(L) := \{\phi : L \rightarrow L \mid \phi|_k \text{ is identity}\}.$$

(a) Let $\alpha : L \rightarrow L$ a k -homomorphism. Show that $\alpha \in \text{Aut}_k(L)$. [2]

(b) Let $f(x)$ has m distinct roots in L . Then construct a injective group homomorphism $\theta : \text{Aut}_k(L) \rightarrow S_m$ such that $\text{Image}(\theta) \subset S_m$ is a transitive subgroup. [5]

Handwritten calculations:

$$\begin{array}{r} 27 \\ \times 12 \\ \hline 54 \\ 27 \times \\ \hline 324 \end{array}$$

Other scribbles: 8×10^{-1} , 8×10^{-1} , 8×10^{-1} , 8×10^{-1}